

Probability and Random Processes

Random walks. Exercises

3. Let $\{X_n : n \geq 0\}$ be a random walk with parameter $p = 3/4$.
- (a) What is the probability of a return to the origin at time k ?
 - (b) If at time n we have $X_n = a$, what is the probability of visiting again a after k steps?
 - (c) Calculate the expected number of visits to the origin during the interval $1 \leq n \leq 6$. What is this number if $p = 1/4$?

Solution

- (a) The probability of a return to the origin at time k is 0 if k is odd, and if k is even, then

$$\mathbb{P}(X_k = 0) = \binom{k}{k/2} p^{k/2} q^{k/2}.$$

- (b) Since the steps of a random walk are independent, once we reach position a at time n , we have

$$\mathbb{P}(X_{n+k} = b \mid X_n = a) = \mathbb{P}(X_k = b \mid X_0 = a) = \mathbb{P}(X_k = b - a \mid X_0 = 0) = \mathbb{P}(X_k = b - a).$$

In particular,

$$\mathbb{P}(X_{n+k} = a \mid X_n = a) = \mathbb{P}(X_k = 0),$$

which has been computed in part (a).

- (c) Let I_k be the indicator random variable for the event “the walk visits the origin at time k ”, so that $\mathbb{E}(I_k) = \mathbb{P}(X_k = 0)$. Since this probability is zero when k is odd, we can conclude that the expected number of visits to the origin, for $1 \leq k \leq 6$, is given by:

$$\begin{aligned} \mathbb{E}\left(\sum_{i=1}^6 I_i\right) &= \sum_{i=1}^6 \mathbb{E}(I_i) = \sum_{i=1}^6 \mathbb{P}(X_i = 0) \\ &= \mathbb{P}(X_2 = 0) + \mathbb{P}(X_4 = 0) + \mathbb{P}(X_6 = 0) = \binom{2}{1} pq + \binom{4}{2} p^2 q^2 + \binom{6}{3} p^3 q^3. \end{aligned}$$

In particular, if $p = 3/4$, we get:

$$\mathbb{E}\left(\sum_{i=1}^6 I_i\right) = 2 \cdot \frac{3}{4^2} + 6 \cdot \frac{9}{4^4} + 20 \cdot \frac{27}{4^6} = \frac{735}{1024} \approx 0.7178.$$

By symmetry, we obtain the same value for $p = 1/4$.

4. Let $\{X_n : n \geq 0\}$ be a symmetric random walk. Calculate the following probabilities:

- (a) That the random walk does not visit the coordinate -1 for any $1 \leq n \leq 2m$.
- (b) That the random walk does not visit the coordinate -2 for any $1 \leq n \leq 2m$.
- (c) That $X_n \leq 2$ for all $1 \leq n \leq 2m$,

Solution

- (a) The probability α_m that the random walk does not visit the coordinate -1 during the first $2m$ steps can be expressed as:

$$\alpha_m = \mathbb{P}(X_1 \geq 0, \dots, X_{2m} \geq 0).$$

Let us prove that α_m is also equal to $\mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0)$. To see this, consider the following facts:

(a.1) If $X_{2m-1} \geq 0$, then $X_{2m-1} > 0$, and so $X_{2m} \geq 0$. Therefore,

$$\mathbb{P}(X_1 \geq 0, \dots, X_{2m-1} \geq 0, X_{2m} \geq 0) = \mathbb{P}(X_1 \geq 0, \dots, X_{2m-1} \geq 0).$$

(a.2) By the homogeneity of the random walk, and taking into account that $X_0 = 0$,

$$\begin{aligned} \mathbb{P}(X_1 \geq 0, \dots, X_{2m-1} \geq 0) &= \mathbb{P}(X_1 \geq 0, \dots, X_{2m-1} \geq 0 \mid X_0 = 0) \\ &= \mathbb{P}(X_2 \geq 1, \dots, X_{2m} \geq 1 \mid X_1 = 1) \\ &= \mathbb{P}(X_2 > 0, \dots, X_{2m} > 0 \mid X_1 = 1). \end{aligned}$$

(a.3) Moreover,

$$\begin{aligned} \mathbb{P}(X_2 > 0, \dots, X_{2m} > 0 \mid X_1 = 1) &= \frac{\mathbb{P}(X_1 = 1, X_2 > 0, \dots, X_{2m} > 0)}{\mathbb{P}(X_1 = 1)} \\ &= 2 \mathbb{P}(X_1 > 0, X_2 > 0, \dots, X_{2m} > 0). \end{aligned}$$

(a.4) Furthermore, by the symmetry of the random walk,

$$\mathbb{P}(X_1 > 0, \dots, X_{2m} > 0) = \mathbb{P}(X_1 < 0, \dots, X_{2m} < 0).$$

Hence,

$$\begin{aligned} 2 \mathbb{P}(X_1 > 0, \dots, X_{2m} > 0) &= \mathbb{P}(X_1 > 0, \dots, X_{2m} > 0) + \mathbb{P}(X_1 < 0, \dots, X_{2m} < 0) \\ &= \mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0). \end{aligned}$$

Then, putting all together,

$$\begin{aligned} \alpha_m &= \mathbb{P}(X_1 \geq 0, \dots, X_{2m} \geq 0) = \mathbb{P}(X_1 \geq 0, \dots, X_{2m-1} \geq 0) \\ &= \mathbb{P}(X_2 > 0, \dots, X_{2m} > 0 \mid X_1 = 1) = 2 \mathbb{P}(X_1 > 0, X_2 > 0, \dots, X_{2m} > 0) \\ &= \mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0), \end{aligned}$$

as we wanted to prove.

Finally, by using the known fact that $\mathbb{P}(X_1 \neq 0, \dots, X_{2m} \neq 0) = \mathbb{P}(X_{2m} = 0)$ we get

$$\alpha_m = \binom{2m}{m} \frac{1}{2^{2m}}.$$

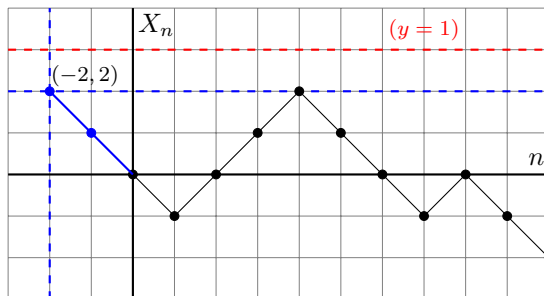
(b) By the spatial and temporal homogeneity of the random walk, and taking into account that if $X_{2m+1} \geq 0$, then $X_{2m+2} \geq 0$, we deduce (reasoning as in the previous item) that:

$$\begin{aligned} \mathbb{P}(X_1 \geq -1, \dots, X_{2m} \geq -1) &= \mathbb{P}(X_2 \geq 0, \dots, X_{2m+1} \geq 0 \mid X_1 = 1) \\ &= 2 \mathbb{P}(X_1 \geq 0, X_2 \geq 0, \dots, X_{2m+1} \geq 0) \\ &= 2 \mathbb{P}(X_1 \geq 0, X_2 \geq 0, \dots, X_{2m+1} \geq 0, X_{2m+2} \geq 0) \\ &= 2 \alpha_{m+1} = \binom{2m+2}{m+1} \frac{1}{2^{2m+1}} = \binom{2m+1}{m} \frac{1}{2^{2m}}, \end{aligned}$$

where in the last equality, we have used that

$$\binom{2m+2}{m+1} = \binom{2m+1}{m} + \binom{2m+1}{m+1} = \binom{2m+1}{m} + \binom{2m+1}{m} = 2 \binom{2m+1}{m}$$

(c) Let us consider the (n, X_n) -plane representation of the random walk. By placing the origin at $(-2, 2)$, we observe that the probability that $X_n \leq 2$ for $n = 1, 2, \dots, 2m$ corresponds to the probability that a path of length $2m + 2$ does not touch the line $y = 1$, given that its first two steps are both “down”.



In other words, if $X'_1 = -1$, $X'_2 = -2$, and $X'_n = X_{n-2} - 2$ for $n \geq 3$, then

$$\begin{aligned}\mathbb{P}(X_1 \leq 2, \dots, X_{2m} \leq 2) &= \mathbb{P}(X'_1 = -1, X'_2 = -2, X'_3 \leq 0, \dots, X'_{2m+2} \leq 0 \mid X'_1 = -1, X'_2 = -2) \\ &= 4 \mathbb{P}(X'_1 = -1, X'_2 = -2, X'_3 \leq 0, \dots, X'_{2m+2} \leq 0) \\ &= 4 \mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leq 0, \dots, X_{2m+2} \leq 0),\end{aligned}\tag{1}$$

where the last equality is because $\{X_n: n \geq 1\}$ and $\{X'_n: n \geq 1\}$ are both symmetric random walks. Moreover, notice that if α_m is defined as in part (a), by the symmetry of the random walk, we have

$$\begin{aligned}\alpha_{m+1} &= \mathbb{P}(X_1 \leq 0, X_2 \leq 0, \dots, X_{2m+2} \leq 0) \\ &= \mathbb{P}(X_1 = -1, X_2 = 0, X_3 \leq 0, \dots, X_{2m+2} \leq 0) + \mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leq 0, \dots, X_{2m+2} \leq 0).\end{aligned}$$

Thus, since

$$\begin{aligned}\mathbb{P}(X_1 = -1, X_2 = 0, X_3 \leq 0, \dots, X_{2m+2} \leq 0) &= \mathbb{P}(X_1 = -1, X_2 = 0) \mathbb{P}(X_3 \leq 0, \dots, X_{2m+2} \leq 0 \mid X_2 = 0, X_1 = 0) \\ &= \frac{1}{4} \mathbb{P}(X_3 \leq 0, \dots, X_{2m+2} \leq 0 \mid X_2 = 0) \\ &= \frac{1}{4} \mathbb{P}(X_1 \leq 0, \dots, X_{2m} \leq 0 \mid X_0 = 0) = \frac{1}{4} \mathbb{P}(X_1 \leq 0, \dots, X_{2m} \leq 0) \\ &= \frac{1}{4} \alpha_m,\end{aligned}$$

and, by (1),

$$\mathbb{P}(X_1 = -1, X_2 = -2, X_3 \leq 0, \dots, X_{2m+2} \leq 0) = \frac{1}{4} \mathbb{P}(X_1 \leq 2, \dots, X_{2m} \leq 2),$$

we can conclude that

$$\alpha_{m+1} = \frac{1}{4} \alpha_m + \frac{1}{4} \mathbb{P}(X_1 \leq 2, \dots, X_{2m} \leq 2).$$

Finally, we obtain that:

$$\begin{aligned}\mathbb{P}(X_1 \leq 2, \dots, X_{2m} \leq 2) &= 4\alpha_{m+1} - \alpha_m \\ &= \left(\binom{2m+2}{m+1} - \binom{2m}{m} \right) \frac{1}{2^{2m}} = \frac{3m+1}{m+1} \binom{2m}{m} \frac{1}{2^{2m}}.\end{aligned}$$

6. In an election, there are two candidates, A and B , and each of $2m$ voters votes with equal probability for A or B .

- If candidate A has received two votes more than candidate B , what is the probability that A has always been in the lead during the counting of the votes?
- What is the answer to the previous question if A has received $2r$ votes more than B ?
- What is the probability that one of the candidates has always been in the lead while the votes are being counted?
- If candidate A has never fallen behind candidate B during the counting of the votes, calculate the probability that the counting finishes with an equal number of votes for both candidates.

Solution

Each counting of the votes can be understood as a path in the (n, X_n) -plane representation, where a step “up” represents a vote for candidate A , while a step “down” signifies a vote for candidate B . Our hypothesis is that all potential paths connecting the starting point $(0, 0)$ to the final result at $(2m, 2k)$, $-m \leq k \leq m$, have an equal probability of occurrence. We will make use of the reflection principle.

- If A has received two votes more than B , the possible trajectories of the random walk correspond to paths from $(0, 0)$ to $(2m, 2)$. Since each of these paths consists of $m-1$ “down” steps and $m+1$ “up” steps, the total number is $\binom{2m}{m+1}$.

If candidate A must always be in the lead (which means that the number of votes for A is always greater than the number of votes for B), then the random walk must always remain strictly positive. In other words, $X_n > 0$ for all $1 \leq n \leq 2m$. The number of valid paths from the point $(0, 0)$ to $(2m, 2)$ that meet this criterion is equal to the number of paths with the same property from the point $(1, 1)$ to $(2m, 2)$. This number is given by $\binom{2m-1}{m}$ minus the number of paths from $(1, 1)$ to $(2m, 2)$ that touch or cross the horizontal n -axis. By applying the reflection principle, we can find that the latter quantity is equivalent to the number of paths from $(1, -1)$ to $(2m, 2)$, which is $\binom{2m-1}{m+1}$.

Thus, the probability that candidate A has consistently been in the lead during the counting of the votes, given that A ends up with two more votes than B , is:

$$\frac{\binom{2m-1}{m} - \binom{2m-1}{m+1}}{\binom{2m}{m+1}} = \frac{1}{m}.$$

- (b) Reasoning as in part (a), we deduce that if A has received $2r$ votes more than B , then the probability that during the counting of the votes A has always been in the lead is:

$$\frac{\binom{2m-1}{m+r-1} - \binom{2m-1}{m+r}}{\binom{2m}{m+r}} = \frac{r}{m}.$$

- (c) The probability that one of the candidates is always in the lead during the counting of the votes is the probability that a symmetric random walk does not return to the origin within the first $2m$ steps, which is also equal to:

$$\mathbb{P}(X_{2m} = 0) = \binom{2m}{m} \frac{1}{2^{2m}}.$$

- (d) By applying the result from Problem 4(a), we find that the total number of trajectories of a symmetric random walk that satisfies $X_1 \geq 0, \dots, X_{2m} \geq 0$ (where candidate A has never fallen below candidate B) is given by $\binom{2m}{m}$.

On the other hand, the number of these trajectories that conclude with an equal number of votes for both candidates corresponds to the number of paths from $(0, 0)$ to $(2m, 0)$ that do not touch the horizontal line $y = -1$. Utilizing the reflection principle, we can determine that this number is represented by the Catalan number:

$$C_m = \binom{2m}{m} - \binom{2m}{m+1} = \frac{1}{m+1} \binom{2m}{m}.$$

Consequently, if candidate A has never been below candidate B , the probability that the counting concludes with an equal number of votes for both candidates is given by the ratio:

$$\frac{\frac{1}{m+1} \binom{2m}{m}}{\binom{2m}{m}} = \frac{1}{m+1}.$$